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Delhi Public School, Howrah

PRE-BOARD I EXAMINATION (2025-2026)

Class-X

Care must be taken not to write anything on the question paper. All the questions must be attempted in the correct sequence.

Subject: - Science (Code No. 086)

F.M.-80

Time:- 3 Hours

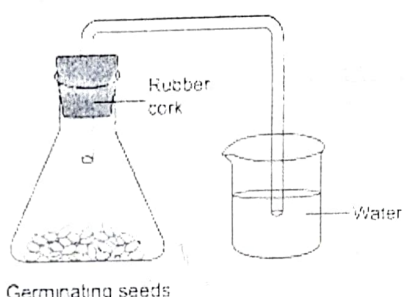
General Instructions:

- This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.
- All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.

SECTION - A

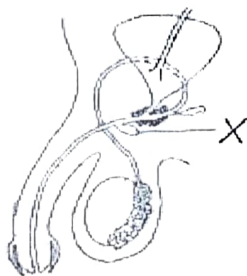
- Which of these reactions occur in photosynthesis? 1
 - Carbon dioxide is reduced, and water is oxidized.
 - Water is reduced and carbon dioxide is oxidized.
 - Both carbon dioxide and water are oxidized.
 - Both carbon dioxide and water are reduced.
- The following experiment was set up to show that a gas is given out during respiration. But there was no rise in the level of water. This was because- 1

- Germinating seeds have not been kept under water in the flask.
 - Water is kept in the beaker instead of lime water.
 - The cork on the flask is made of rubber.
 - No substance is kept in the flask to absorb the gas given out by the seeds.



Germinating seeds
- The correct sequence of anaerobic respiration - 1
 - Glucose → pyruvate → lactic acid
 - Glucose → pyruvate → carbon dioxide + ethanol
 - Glucose → pyruvate → carbon dioxide + ethanol + energy
 - Glucose → pyruvate → ADP → lactic acid
- Hydra reproduces by budding. The new individual formed will - 1
 - Be genetically different from the parent.
 - Have half the genes of the parent.
 - Be genetically identical to the parent.
 - Show new variations.
- If H represents for tall trait which is dominant and h is the recessive trait for short, which of the following cross will result in 1:1 tall: short progeny? 1
 - HH X hh
 - Hh X Hh
 - Hh X hh
 - hh X hh

6. The diagram given below shows a section of the male reproductive system. What is the function of part X? 1



- A. The secretion of the gland neutralizes acids from urine.
 B. It decreases the number of sperms damaged during ejaculation.
 C. It stimulates uterine contraction to help sperms move forward into the fallopian tube.
 D. Both (A) and (B)
7. In the given food chain if the amount of energy at the fourth trophic level is 4 kJ, what will be the energy available at the producer level? 1
 Grass → Grasshopper → Frog → Snake
- A. 4 kJ
 B. 40 kJ
 C. 400 kJ
 D. 4000 kJ

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

- A. Both A and R are true, and R is the correct explanation of A.
 B. Both A and R are true, but R is not the correct explanation of A.
 C. A is true but R is false.
 D. A is false but R is true.

8. **Assertion (A):** The ratio of tall to dwarf plants in Mendel's F_2 generation is 3:1. 1
Reason (R): The F_1 tall plants are heterozygous, and their self-pollination produces both tall and dwarf plants.

9. **Assertion (A):** Animals can react to stimuli in different ways. 1
Reason (R): All animals have a nervous system and an endocrine system involving hormones.

10. What would be the ratio of chromosome numbers between an egg and a zygote? How is the sperm genetically different from the egg? 2

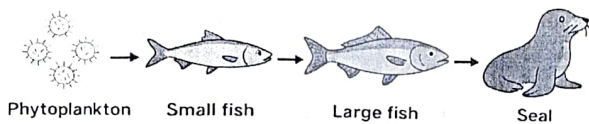
11. Students to attempt either option A or B. 2

A. In pea plants, green pods are dominant over yellow pods. Cross a plant with yellow pods with another plant that is heterozygous for green pods. (Show the cross by choosing the correct alphabet for the allele of pod colour)

OR

B. In pea plants, the seed coat can be green or white. Green is dominant over white. Construct your own Punnett square in which 50% of the offspring can be predicted to have green seed coats. (Choose the correct alphabet for the seed coat colour).

12. A. If only 1% of the sunlight reaching Earth is trapped by plants, explain how the 10% law affects the length and stability of food chains in ecosystems. 2
 B.



In the above food chain if the energy at the producer level is 40,000 kJ, find the energy available to seals.

13. A. A person accidentally touches a hot iron rod and immediately withdraws his hand. Later, he feels the pain. Why does the withdrawal happen before feeling the pain? 3
 B. Farmers often pinch the tips of young plants to promote lateral growth. Explain scientifically why this is done. 0.5
 C. Why are electrical impulses not suitable for controlling growth and development in plants? 0

14. Raghav was watching Nia and made the following observation:
 Nia was practicing boxing. As her opponent swung an arm, Nia blinked her eyes and ducked down. Nia retaliated with a punch.
 A. Raghav said that the blinking of eyes and ducking down are both reflex actions. Is he correct? Justify.
 B. Explain the nervous process involved in Nia's retaliation with a punch in TWO points. 3

15. Read the following case and answer the following questions: 4

In a pea plant species, yellow seed colour (Y) is dominant over green (y), and round seed shape (R) is dominant over wrinkled (r).

A farmer plants 100 pea seeds obtained from self-pollinated F_1 plants of genotype YyRr.

- A. How many of the plants are expected to show both dominant traits?
 B. How many would show both recessive traits?

Attempt either subpart C or D.

- C. What law predicts this pattern of inheritance? If the farmer selects only the yellow round seeds for the next sowing, which type of cross will he perform?

OR

- D. If instead of 100, the farmer planted 160 seeds from the same F_1 generation (YyRr), calculate the expected number of each phenotype in F_2 .

16. Attempt either option A or B.

2 + 3

- A. I. There are different nutrients required by the human body. These are in 3 major categories of carbohydrates, fats and proteins (apart from vitamins, minerals and roughage).
 (i) Digestion of nutrient R happens in the stomach. Identify R.
 (ii) Consider fats and oils from your diet as large globules in your digestive tract. Name the reaction that can help in making these easier to absorb.
 (iii) What is the difference in the kind of medium required for digestive enzymes in the stomach and the small intestine to work?

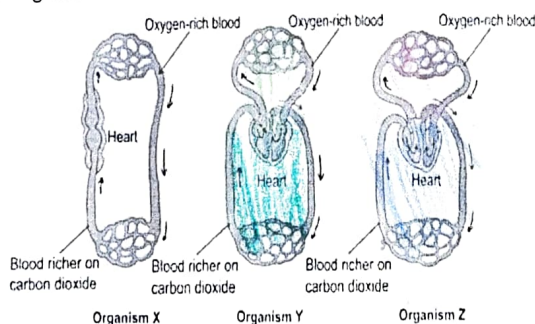
- II. Draw the following diagrams with proper labelling: [1+2]

- (i) Nutrition in *Amoeba*
 (ii) Cross-section of a leaf

OR

- B. I. (i) In respiration, glucose is broken down into a 3-C molecule called pyruvate, in the cytoplasm of the cell. Is this particular process aerobic or anaerobic in nature? Justify your answer.

- (ii) Given below are the diagrammatic representations of the circulatory systems of three organisms X, Y and Z. Identify which organism is a fish. Give a reason for your answer.

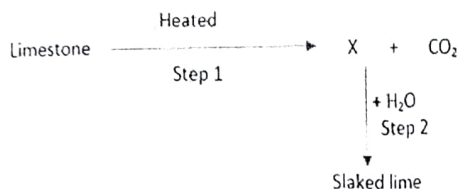


- II. Draw the following diagrams with proper labelling: [2+1]

- (i) Schematic diagram of human heart.
 (ii) Movement of water during transpiration in trees.

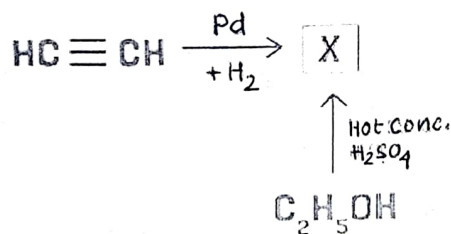
SECTION - B

17. The electronic configuration of three elements X, Y and Z are X = 2, 8, 7; Y = 2, 8, 2; and Z = 2, 8. Choose the correct answer from the given information. 1
- A. Y and Z are metals
 B. Y and X are non-metals
 C. X is a non-metal and Y is a metal
 D. Y is a non-metal and Z is a metal
18. During a science exhibition, a student dropped a small piece of sodium into water. A vigorous reaction took place with the release of a gas that caught fire spontaneously. Which statement best explains the observation? 1
- A. Sodium reacts endothermically with water.
 B. Sodium reacts with water forming hydrogen gas and sodium hydroxide.
 C. Sodium reacts with water forming oxygen gas and sodium oxide.
 D. Sodium absorbs hydrogen gas from water.
19. Identify the correct option from the given table which represents the type of reactions occurring in step 1 and step 2 respectively as shown below. 1



	Endothermic	Exothermic
A.	X	✓
B.	✓	X
C.	✓	✓
D.	X	X

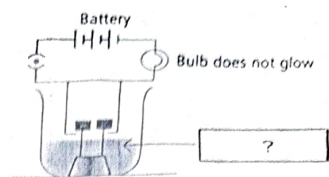
20. Two hydrocarbons, X and Y, have molecular formulae C_2H_4 and C_2H_6 respectively. When both are passed through bromine water, only one causes the solution to decolourise. Which of the two hydrocarbons is unsaturated? 1
- A. X
 B. Y
 C. Both X & Y.
 D. Cannot be concluded.
21. Ipsita added a drop of diluted acetic acid and diluted hydrochloric acid on pH paper and compared the colors. Select the correct conclusion based on pH range of indicators. 1
- A. pH of acetic acid is more than that of hydrochloric acid.
 B. pH of acetic acid is less than that of hydrochloric acid.
 C. Acetic acid dissociates completely in aqueous solution.
 D. Acetic acid is a strong acid
22. Which of the following correctly represents 'X' in the given chemical changes? 1



- A. Ethane
 B. Ethene
 C. Ethyne
 D. Ethanoic acid

23. The solution in the given figure is likely to be

- A. HNO_3
 B. $\text{C}_2\text{H}_5\text{OH}$
 C. H_2SO_4
 D. CO_2 in water



1

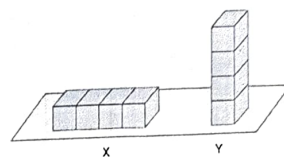
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 B. Both A and R are true, and R is not the correct explanation of A.
 C. A is true, but R is false.
 D. A is false, but R is true.

24. Assertion (A): Aluminium develops a thin layer of aluminium oxide when exposed to air, preventing further corrosion.

Reason (R): Aluminium oxide is a protective, non-reactive layer that prevents air and moisture from reaching the metal surface

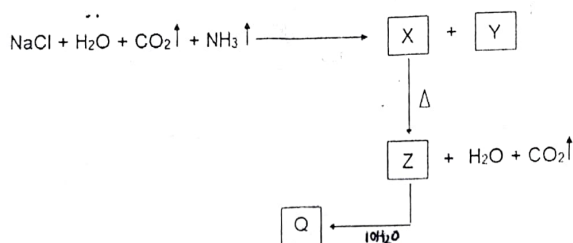
25. A. Which metal can be extracted from cinnabar?
 B. Eight identical, iron blocks are placed on the ground in the two arrangements X and Y as shown alongside. The block arrangements are kept moist by sprinkling water every few hours. Which of the arrangements is likely to gather more rust after ten days? Justify your answer.



1

2

26.



3

- A. Identify compounds X, Y, Z, and Q in the above reaction scheme.
 B. If compound X is heated strongly without the presence of Y, which gas(es) would be evolved?

27. Attempt either option A or B.

3

- A. pH is a key factor to determine the strength of acid and base in a solution. The value of pH changes according to H_3O^+ concentration in a solution. Based on the fact, answer the following questions.
 I. Tooth enamel gets corroded when pH in the mouth falls below 5.5. Why does this happen?
 II. What happens to the pH if an acidic solution is diluted?
 III. During a titration experiment, a student notices that the colour of phenolphthalein changes from pink to colourless when an unknown solution is added dropwise. What can be inferred about the change in pH in this case?

OR

- B. Vinegar is often used to clean tarnished copper vessels.

- I. What causes the tarnishing of copper?
 II. Write the type of reaction taking place when vinegar is applied to it.
 III. Why should the cleaned vessel be washed with water immediately after applying vinegar?

28. A quiz competition was organized in a school where the quiz master provided certain hints to the participants. The participants were expected to interpret the clues carefully and answer accordingly. Based on the hints given by the quiz master, answer the following questions.
 (i) Substance 'C' is used as a preservative.

4

- (ii) 'C' has two carbon atoms; 'C' is obtained by the reaction of 'A' in presence of alkaline Potassium permanganate followed by acidification.
- (iii) Misuse of 'A' in industries is prevented by adding methanol, benzene, and pyridine to 'A'.
- (iv) 'F' is formed on heating 'A' in presence of conc. Sulphuric acid.
- (v) 'F' reacts with Hydrogen gas in presence of Nickel and Palladium catalyst.

A. Misuse of substance 'A' is prevented by adding methanol, benzene, and pyridine because –

(a) It increases purity of 'A' (b) It makes 'A' poisonous and unfit for drinking

(c) It changes 'A' to 'C' (d) It helps in the oxidation of 'A'

Attempt either part B or C.

B. Identify substances 'A', 'C', and 'F' and write the sequence of reactions connecting them.

OR

C. How can you distinguish experimentally between 'A' and 'F'?

D. Heating substance 'A' with conc. H_2SO_4 produces 'F'. The reaction involves –

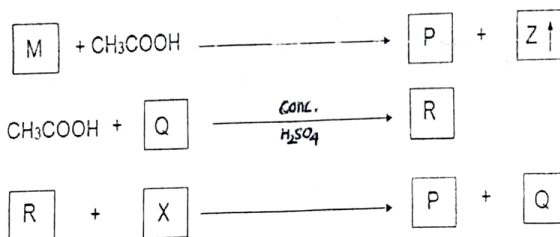
(a) Dehydration (b) Dehydrogenation (c) Oxidation (d) Substitution

29. Attempt either option A or B.

- A. I. Why is methane considered a cleaner fuel compared to coal?
- II. What happens when ethanol is heated with concentrated H_2SO_4 at $170^\circ C$?
- III. Why cannot vegetable oil be converted into ghee by heating or cooling?
- IV. Show the electron dot structure of ethane.
- V. Intake of small quantities of methanol can be lethal. Comment.

OR

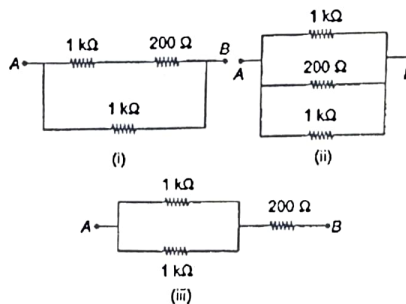
B. Analyze the reaction scheme given and answer the following.



- I. What is the IUPAC name of compound P?
- II. How can you identify the gas Z that has evolved in the first reaction.
- III. What type of reaction occurs when R reacts with X to form P and Q?
- IV. What is the role of concentrated H_2SO_4 in the second reaction?
- V. What is the functional group present in compound R?

SECTION - C

30. In which of the following networks of resistors as shown alongside the equivalent resistance between points A and B is maximum?



- A. Network (i)
- B. Network (ii)
- C. Network (iii)
- D. All have equal equivalent resistance.

31. The side of the Moon that is visible to us from Earth is called the near side while the side that is invisible to us is called the far side. What colour would the sky appear to an astronaut standing on the "far side" of the Moon and why? 1

- A. Blue, as the Moon's atmosphere scatters sunlight just like Earth
- B. White, as the Moon's surface reflect all the light that falls on it
- C. Black, as there is no atmosphere on Moon to scatter sunlight
- D. Black, as sunlight does not fall on the far side of the Moon

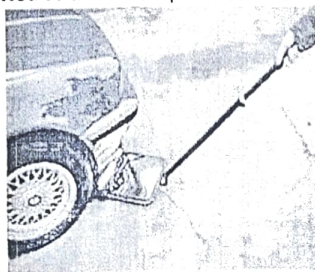
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- D. A is false but R is true.

32. **Assertion(A):** A compass needle is placed near a current carrying wire. The deflection of the compass needle decreases when the magnitude of the current in the wire is increased. 1

Reason(R): The strength of a magnetic field at a point near the conductor increases on increasing the current.

33. Search mirrors are mirrors that are used to look for hidden objects underneath the cars as shown. The hidden objects can be easily spotted as the mirror provides a wider field of view. 2



- A. What type of mirrors are generally used to make search mirrors?
- B. With the help of a ray diagram describe the nature of image formed when an object is kept at any finite distance from the mirror identified in Part A.

34. Attempt either option A or B. 2

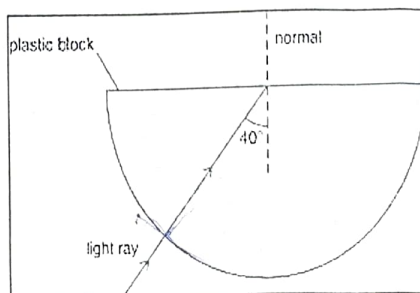
- A. Two wires of equal length, one of copper and the other of manganin (an alloy) have the same thickness. Which one can be used for (i) electrical transmission, (ii) electrical heating devices? 2
- Give proper reason.

OR

- B. A piece of wire of resistance R is cut into three equal parts. These parts are then connected in parallel. If the equivalent resistance of this parallel combination is R_1 , what is the value of the ratio $R_1 : R$? 2

35. A. Explain why the refractive index of any material with respect to air is always greater than 1. 3

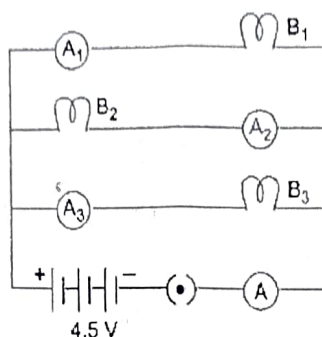
- B. In the figure alongside a light ray travel from air into the semi-circular plastic block. Give a reason why the ray does not deviate at the semi-circular boundary of the plastic block.



- C. Complete the ray diagram of the above scenario when the light ray comes out of the plastic block from the top flat end.

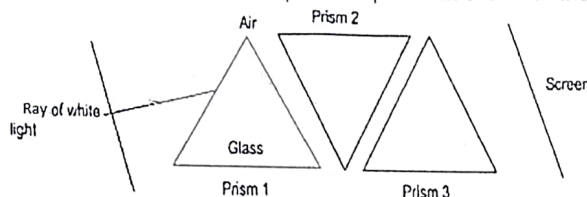
36. B_1 , B_2 , and B_3 are three identical bulbs connected as shown in the given figure. When all the three bulbs glow together, a current of 3A is recorded by the ammeter A.

3



- A. What happens to the intensity of the other two bulbs when the bulb B_1 gets fused? Justify it.
 B. What will be the readings of ammeters A_1 , A_2 , A_3 and A when the bulb B_2 gets fused?
 C. How much power is dissipated in the circuit when all the three bulbs glow together? Show proper calculations.
37. A beam of white light passes through a series of equilateral prisms as shown in the figure below.

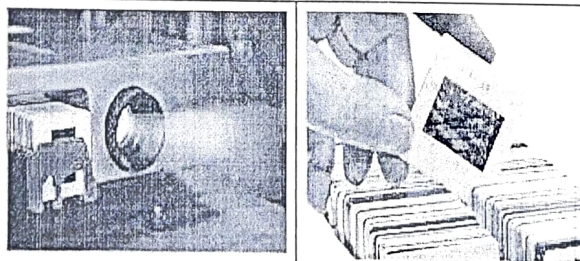
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- A. What colour/(s) will be seen on the screen?
 B. Copy the given diagram above and show the path of the ray of light as it enters Prism 1, passes through the Prisms and emerges from Prism 3 to fall on the screen.
 C. Name all the processes that take place when the ray of light enters Prism 1 and emerges from Prism 3.
38. The given images are that of a specialized slide projector. Slides are small transparencies mounted in sturdy frames ideally suited for magnification and projection. Since they have a very high resolution and a high image quality. There is a tray where the slides are to be put in a particular orientation so that the viewers can see the enlarged erect images of the transparent slides. This means that the slides will have to be inserted upside down in the projector tray.

4

To show her students the images of insects that she investigated in the lab, Mrs. Sumana brought a slide projector. Her slide projector produced 500 times enlarged and inverted image of a slide on a screen 10 m away.



- A. Based on the information given in the above paragraph, what kind of lens must the slide projector have?
 B. If v is the symbol used for image distance and u for object distance then with one reason state what will be the sign for v/u in the given case?

Attempt either subpart C or D.

- C. A slide projector has a convex lens with a focal length of 20 cm. The slide is placed upside down 21 cm from the lens. How far away should the screen be placed from the slide projector's lens so that the slide is in focus?

OR

- D. When a slide is placed 15 cm behind the lens in the projector, an image is formed 3 m in front of the lens. If the focal length of the lens is 14 cm, draw a ray diagram to show image formation. (not to scale)

39. Attempt either option A or B.

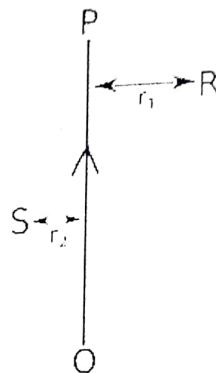
- A. PQ is a current carrying conductor in the plane of the paper as shown in the given figure.

(i) Find the directions of the magnetic fields produced by it at points R and S.

(ii) Given $r_1 > r_2$, where will the strength of the magnetic field be larger? Give reasons.

(iii) If the polarity of the battery connected to the wire is reversed, how would the directions of the magnetic field be changed?

(iv) Explain the rule that is used to find the direction of the magnetic field for a straight current carrying conductor.



(1+2

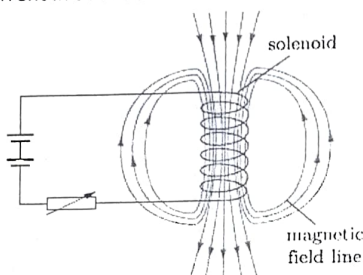
+1

+1)

OR

- B. (a) A circuit contains a battery, a variable resistor and a solenoid. The figure below shows the magnetic field pattern produced by the current in the solenoid.

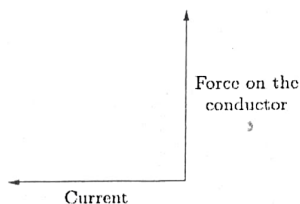
(2+3)



(i) State how the magnetic field pattern indicates regions where the magnetic field is stronger?

(ii) What happens to the magnetic field when the current in the circuit is reversed?

(b) A conductor is placed in a magnetic field and when current passes through the conductor, it experiences a force due to the magnetic field as depicted in the figure below.



(i) State the direction of magnetic field in the case shown above.

(ii) Also state the rule you have used here.